

20260916 Quiz

for (MATH200) Differential Equations

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Problem 2.1.3. Find the general solution of the differential equation

$$y' + y = te^{-t} + 2,$$

and use it to determine how solutions behave as $t \rightarrow \infty$.

Solution. The integrating factor is

$$\mu(t) = \exp\left(\int 1 \, dt\right) = e^t.$$

Multiplying the equation by e^t gives

$$(e^t y)' = t + 2e^t.$$

Hence

$$e^t y = \frac{t^2}{2} + 2e^t + c,$$

so the general solution is

$$y = 2 + \left(\frac{t^2}{2} + c\right) e^{-t}.$$

Since $t^2 e^{-t} \rightarrow 0$ and $e^{-t} \rightarrow 0$ as $t \rightarrow \infty$, every solution satisfies

$$y(t) \rightarrow 2 \quad \text{as } t \rightarrow \infty.$$

□

Problem 2.1.12. Find the solution of the initial value problem

$$ty' + 2y = 2 \sin t, \quad y\left(\frac{\pi}{2}\right) = 1, \quad t > 0.$$

Solution. Dividing by t gives the standard form

$$y' + \frac{2}{t}y = \frac{2 \sin t}{t}.$$

Since $t > 0$, an integrating factor is

$$\mu(t) = \exp\left(\int \frac{2}{t} \, dt\right) = t^2.$$

Thus

$$(t^2 y)' = 2t \sin t.$$

Integrating by parts gives

$$t^2 y = -2t \cos t + 2 \sin t + c.$$

Using $y(\pi/2) = 1$ yields

$$\frac{\pi^2}{4} = 2 + c, \quad c = \frac{\pi^2}{4} - 2.$$

Therefore

$$y(t) = \frac{-2t \cos t + 2 \sin t + \frac{\pi^2}{4} - 2}{t^2}, \quad t > 0.$$

□

Problem 2.2.5. Solve the differential equation

$$\frac{dy}{dx} = \frac{3x - e^{-x}}{2y + e^y}.$$

Solution. Separating the variables gives

$$(2y + e^y) dy = (3x - e^{-x}) dx.$$

Integrating both sides,

$$y^2 + e^y = \frac{3}{2}x^2 + e^{-x} + c.$$

Equivalently, the solution is given implicitly by

$$2y^2 - 3x^2 + 2e^y - 2e^{-x} = c, \quad 2y + e^y \neq 0.$$

□

Problem 2.2.9. Consider the initial value problem

$$y' = (1 - 2x)y^2, \quad y(0) = -\frac{1}{12}.$$

- (a) Find the solution explicitly.
- (b) Plot the graph of the solution.
- (c) Determine (at least approximately) the interval in which the solution is defined.

Solution. (a) Separating variables gives

$$y^{-2} dy = (1 - 2x) dx.$$

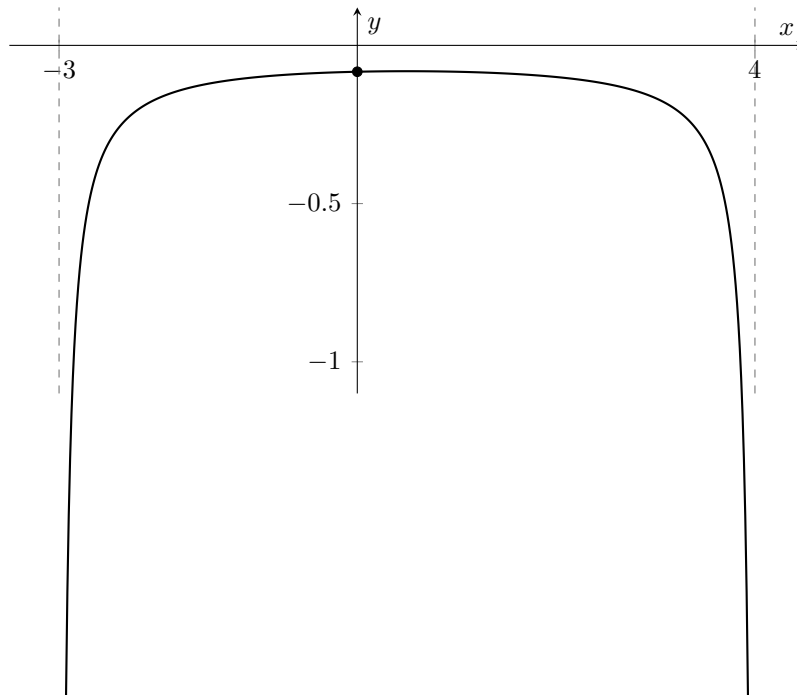
Thus

$$-\frac{1}{y} = x - x^2 + c.$$

The initial condition gives $c = 12$, so

$$y(x) = \frac{1}{x^2 - x - 12} = \frac{1}{(x + 3)(x - 4)}.$$

(b) The graph of the solution branch through $(0, -1/12)$ is shown below. The marked point is the initial point, and the dashed lines indicate the vertical asymptotes $x = -3$ and $x = 4$.



(c) The denominator vanishes at $x = -3$ and $x = 4$. Therefore the largest open interval containing the initial point $x = 0$ on which the solution is defined is

$$\boxed{(-3, 4)}.$$

□

Problem 2.3.1. Consider a tank used in certain hydrodynamic experiments. After one experiment the tank contains 150 L of a dye solution with a concentration of 1 g/L. To prepare for the next experiment, the tank is rinsed with fresh water flowing in at a rate of 2 L/min, while the well-stirred solution flows out at the same rate. Find the time that will elapse before the concentration of dye in the tank reaches 1% of its original value.

Solution. Let $Q(t)$ be the amount of dye in the tank, in grams, at time t minutes. Initially,

$$Q(0) = 150.$$

Since fresh water contains no dye, the dye enters at rate 0. The volume remains 150 L, so the dye leaves at rate

$$\frac{Q(t)}{150} \cdot 2 = \frac{Q(t)}{75} \text{ g/min.}$$

Hence

$$Q' = -\frac{1}{75}Q, \quad Q(0) = 150,$$

and therefore

$$Q(t) = 150e^{-t/75}.$$

The concentration is $Q(t)/150 = e^{-t/75}$ g/L. It reaches 1% of its original value when

$$e^{-t/75} = 0.01.$$

Thus

$$\boxed{t = 75 \ln 100 \text{ min} \approx 345.4 \text{ min}}.$$

□

Problem 2.3.10. Newton's law of cooling states that the temperature of an object changes at a rate proportional to the difference between its temperature and that of its surroundings. Suppose that the temperature of a cup of coffee obeys Newton's law of cooling. If the coffee has a temperature of 90°C when freshly poured, and 1 minute later has cooled to 85°C in a room at 20°C , determine when the coffee reaches a temperature of 65°C .

Solution. Let $T(t)$ be the temperature of the coffee, in degrees Celsius, at time t minutes. Newton's law gives

$$T' = -k(T - 20), \quad k > 0.$$

Thus

$$T(t) = 20 + ce^{-kt}.$$

Since $T(0) = 90$, we have $c = 70$, so

$$T(t) = 20 + 70e^{-kt}.$$

Using $T(1) = 85$ gives

$$e^{-k} = \frac{65}{70} = \frac{13}{14}.$$

When $T(t) = 65$,

$$e^{-kt} = \frac{45}{70} = \frac{9}{14}.$$

Therefore

$$t = \frac{\ln(9/14)}{\ln(13/14)} \text{ min} \approx 5.96 \text{ min}.$$

□